

Experiment 6: Optical tweezers: Measuring piconewton forces

1. Introduction

NOTE: Before attempting this experiment, read this entire lab description.

Optical tweezers instruments use the forces of laser radiation pressure to trap small particles. Using various techniques, these trapped particles can then be manipulated and forces on the objects in the trap can be measured. The forces that such an instrument is capable of measuring are of the order of 0.01 to 300 piconewtons (pN). While this technique has been used for over 20 years to manipulate and study the properties of micron-sized dielectric particles, it is only recently that this precise force measurement instrument has been applied to the study of biological systems.

There are two major types of optical tweezers instruments. In the Williams lab here at Northeastern University, dual-beam optical tweezers instruments are used to measure single molecule DNA-protein interactions. In a dual-beam instrument, two microscope objectives are used to focus two counter-propagating beams to the same spot. The point to which the two beams are focused forms the optical trap. Dual-beam instruments have the advantage that, using a relatively low laser power of 400 mW, they can generate trapping forces of up to 300 pN. The disadvantage of these instruments is that they must be built on an optical table, they involve many optical components, and they require careful alignment.

A single high power laser beam that is focused to a small spot using a high numerical aperture microscope objective will cause small dielectric particles (which in these experiments are usually floating in water) to be attracted to the focus of the beam. This attractive force is due to the refraction of the laser light by the surface of the particle, as shown in Figure 1. For this to work, the index of refraction n of the particle must be greater than that of the surrounding medium. In our experiments we will use polystyrene spheres ($n=1.5$) in water ($n=1.33$).

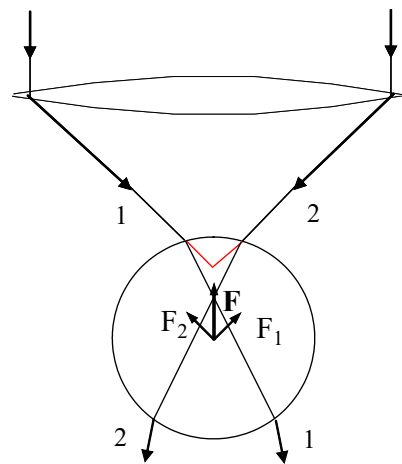
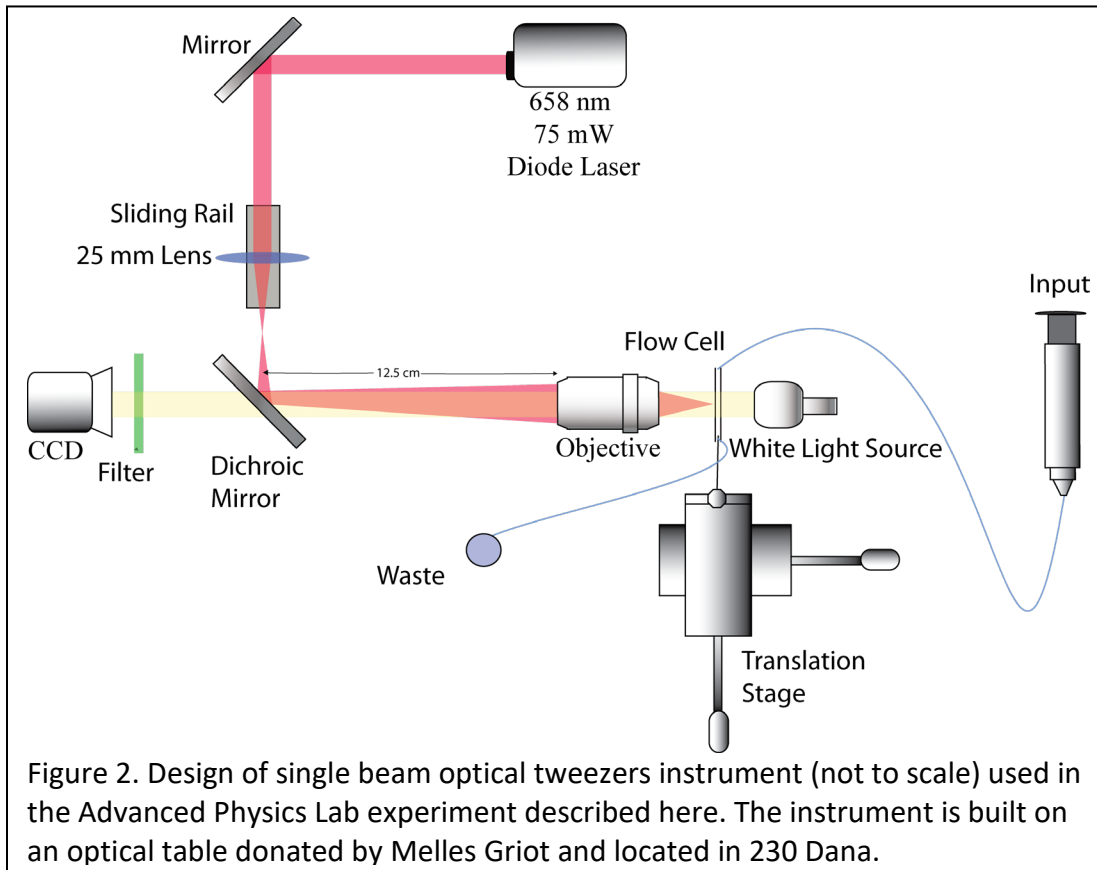


Figure 1. Schematic diagram showing the force on a dielectric sphere due to refraction of two rays of light, 1 and 2. The resultant force on the bead due to refraction is towards the focus.

2. Instrument Design



For our experiments, we will use a single beam optical tweezers design. However, the instrument design is not standard. In fact, most single beam optical tweezers instruments are designed as part of a standard commercial microscope. You will see that a very good microscope can easily be constructed on an optical table with very few optical elements. A diagram of the instrument is shown in Figure 2. Before operating the instrument, be sure to wear the supplied safety glasses, as the 75 mW laser can be dangerous if it is deflected into unprotected eyes. The protective eyewear will reduce the laser intensity by at least two orders of magnitude, rendering even the full 75 mW harmless at 0.75 mW.

Most of the instrument optics will be already set up when you start. However, you will have to set up and mount the flow cell so that you can take data. The flow cell is designed to allow water to flow from into and out of the cell at a fixed rate, such that the liquid flow rate and direction is uniform at the optical trap. Before mounting the flow cell in the instrument, you must fill it with water. The cell consists of a Plexiglas spacer with two glass cover slips glued to each side to form a sealed container with an input and output tube. At the ends of each tube, there is a container with lines on it

every 0.5 ml of liquid volume, as shown in Figure 3. Make sure the clamp screw is tightened so there is no flow. Fill the input container, which has a much longer tube, up to 5 ml with water and mount it on the ring stand. You will use this to fill the cell. Before mounting the cell in the instrument, make sure all of the bubbles are out of the tubing and valve system.

The laser beam is reflected off two laser line mirrors. These mirrors are designed to reflect only light in a relatively narrow band around 633 nm, which includes 675 nm. Between the two mirrors, the beam is expanded by a single convex lens mounted on a sliding rail. The purpose of the lens is to increase the beam diameter so that it fills the

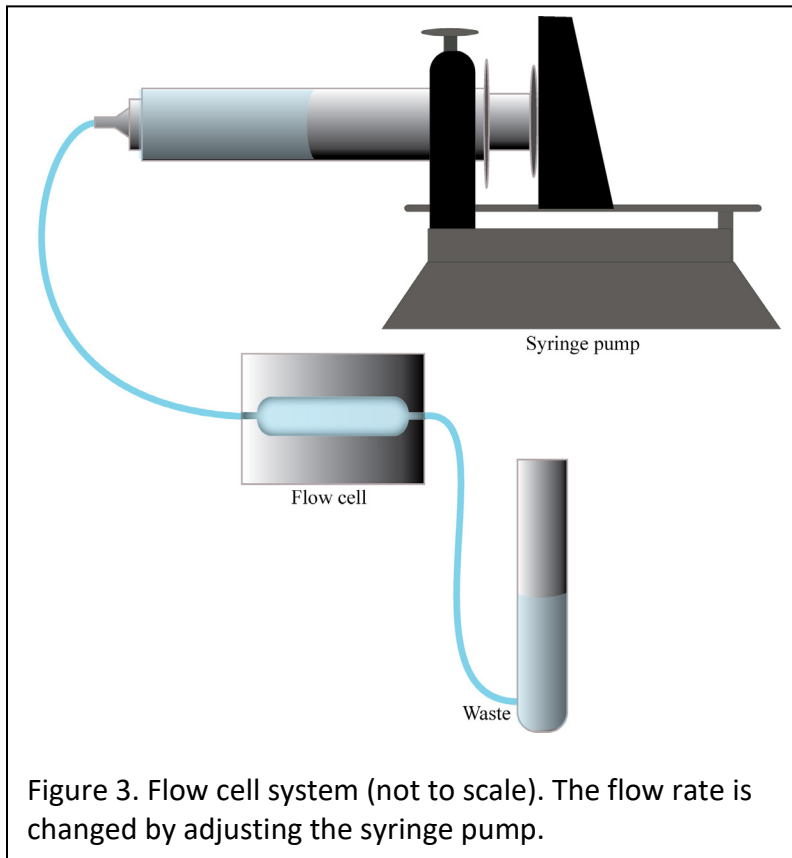


Figure 3. Flow cell system (not to scale). The flow rate is changed by adjusting the syringe pump.

back focal plane of the microscope objective. As you can see from Figure 1, the more illumination you get through the objective, the higher the trapping force will be.

After being reflected off the second mirror, the light is focused by the objective. Then the laser light must be focused into the cell, which we have designed for you. This flow cell must be filled with water and mounted properly very close to the objective. This objective is an “oil immersion” objective, which means you must also have a drop of oil in

between the objective and the cell for it to work properly. After being focused into the cell, the laser beam diverges rapidly.

In order to see what is happening in the cell, we use a light source to create an image. The image comes from looking into the objective, so a camera is placed on the back side of the objective, before the mirror. A blue LED (with dimming knob) is placed directly after the flow cell. This light enters from the front of the objective and most of it goes through the laser line mirror. It is then focused with another lens into the CCD camera. The CCD camera image is displayed on the monitor directly above the table. If a bead is held in the trap, you will see an image on the monitor if the optics are placed properly. Study the optical elements and compare them with what you would expect in a

standard microscope. Which parts correspond to the parts of a microscope? ***Discuss this in your report.***

Because of the geometry in this design, you will not see the laser beam on your camera image. How will you know the beam is focused into the cell? If you use the translation stage to move the cell along the axis of the beam (away from and towards the objective), you should be able to find a place where the laser beam is reflected back towards the camera and you see an image of the focused laser spot. Then move the cell towards the objective using the translation stage for 4 or 5 turns of the micrometer. Thus, you will know the trap is inside the cell and you can trap beads. At this point, you can flow a small amount of bead solution into the cell. As the beads float by, they should be attracted to the focus of the laser beam and they will suddenly come into focus and stay in the trap. Now that you have trapped a bead, you need to figure out how to use it to measure forces.

3. Calibration of the optical tweezers.

The easiest way to measure the force on the bead in the optical trap is to measure its position. The distance of the bead from the center of the trap is directly proportional to the force exerted upon it:

$$\vec{F} = -k\vec{x}, \quad (1)$$

where $\vec{x}$ is the position vector relative to the center of the trap. The parameter k is referred to as the trap stiffness. Once the trap stiffness has been measured, a measurement of the bead position is equivalent to a measurement of the force exerted on the bead. How can we measure this? If we apply a known force and measure the position, we can use linear least-squares fitting to determine k and the error in k .

One common method for applying a known force is to use the viscous drag force on a bead in the trap. The force due to viscous drag on a sphere of known radius can be calculated. If a liquid with viscosity η flows past a sphere of radius r with velocity v , the force due to viscous drag F_{vis} is given by:

$$F_{vis} = \gamma v = 6\pi\eta r v, \quad (2)$$

where γ is the viscous drag coefficient. Since the viscosity of the liquid is known and polystyrene spheres of known radius can be obtained, we can apply a known force if we can measure the velocity of the liquid.

The method we will use for inducing liquid flow is shown in Figure 3. The syringe pump can precisely control the volume flow rate of the liquid. If we knew the cross section of the flow cell precisely and the flow were completely laminar, we could determine the

velocity of the flow past a trapped bead directly for any given volume flow setting. Because we do not have all of this information, here we will measure the flow rate directly by video imaging. The syringe pump will allow you to adjust the flow rate and obtain measurements as a function of flow rate.

To do this, turn on the laser, trap a polystyrene bead, and close the clamp so there is no flow at the moment. Set up the computer to save images of the bead using Pinnacle Studio 24. In the Import tab, you are able to capture video (choose Pinnacle USB-710) and save as an MPEG to a specified location. However, you'll want to have an AVI file for all subsequent analysis. To export as an AVI file, click on the Edit tab where you'll see your most recent captured video. Drag and drop it into the editing track below and click on the Export tab. Now you can save the file as an AVI for further analysis. Please only save a few seconds at a time since we do not have unlimited hard drive space for video capture. Additionally, please delete the original MPEG file since it won't be needed. After capturing an image without flow, pick a flow setting, and turn on the liquid flow by starting the syringe pump. You'll see the displacement of the trapped bead at this point. Now you have the displacement data but you do not know the force. In order to find the force, you'll measure the velocity of a non-trapped bead. That force is proportional to the flow rate at that moment. Record the trapped beads for different flow rates by adjusting the syringe pump flow rate. Make sure to record a non-trapped bead flowing for each trapped instance in order to measure the force and make sure you carefully document the files you are saving under different conditions. You'll notice that flow through the cell is not laminar, but rather that there is a significant flow gradient, such that beads in different planes flow through the cell with different velocities. In order to reduce any systematic error associated with this, try to analyze the velocity of freely flowing beads that are in (or fairly close to) the focal plane (rather than beads that are far out of focus). Obtain measurements of bead displacement with uncertainties at six different flow rates, and make sure these points cover a large range of approximately equally spaced flow rates. In order to determine the uncertainty, you will need at least three measurements at each flow rate (of both the flow rate and the displacement). After analyzing these images and determining the flow rate, you will have a measurement of the trap stiffness. Record the data at the maximum flow rate at which a bead can be trapped in order to estimate the maximum trapping force with uncertainty.

To analyze your images and find the change in position for each velocity, you will can either use National Instruments Vision Assistant or Python. National Instruments Vision Assistant allows you to measure the position of the center of the bead in pixels. Alternatively, you can use the methods developed in Experiment 2 to analyze the images in Python. To find the distance per pixel, measure the size of the bead in pixels, and use the fact that these polystyrene beads are **$4.16 \pm 0.21 \mu\text{m}$ in diameter**. Check with the TA to make sure the beads you are using are described by these numbers. Additionally, when calculating velocity you will need to convert from image frames to time. To do this, use the fact that the camera collects images at a rate of 30 fps. **From**

the data obtained in this experiment, calculate the trap stiffness, maximum trapping force, and the uncertainty in each of these parameters. In your lab write-up, present a few pictures of trapped and freely flowing beads, indicating which functions were used to complete the image analysis. Present a plot of force vs. displacement, including error bars and a fit to the data. Describe how the data are analyzed very clearly, including examples of the measurement from the image. You have now learned to use a powerful modern instrument that is capable of measuring very small forces on single molecules.

NOTE: Please save the bead solution by reusing the waste as input and please do not leave the flow system open to air when you leave. Undergraduate students are not permitted to use the lasers if a TA is not present.

For general information about optical tweezers, see:

Mark C. Williams. 2002. Optical Tweezers: Measuring Piconewton Forces, in *Single Molecule Techniques*, Petra Schwille, ed., a volume of the *Biophysics Textbook Online*, L. DeFelice, editor-in-chief, Biophysical Society, Bethesda, MD. Now available at <https://www.biophysics.org/Portals/0/BPSAssets/Articles/williams.pdf>